

stable. The robustness properties of exponential stability are exploited here to deal with the fast network dynamics, which is therefore treated as a perturbation of the centralized algorithm that can be overcome by a suitably small stepsize. While specialized to handle such a network consensus phenomena, the underlying rationale in which perturbations are handled thanks to the robustness margin of exponential stability can be easily extended to cover other problems of interest, such as time-varying or time-delayed communications, whenever such phenomena can be reduced to fast parasite dynamics that become negligible when the stepsize is sufficiently small. This analysis approach is enabled by the fact that Algorithm (2) employs a non-diminishing stepsize.

As a byproduct of the stability analysis, an explicit theoretical upper bound for the stepsize to guarantee semiglobal linear convergence is provided. Such a bound highlights how the network size, structure, and connectivity, the Lipschitz constant, and the convexity properties of the involved functions influence the stepsize and the convergence rate. Finally, we remark that, for simplicity of exposition, a single stepsize for both the primal and the dual updates is assumed. While this is generally a more difficult case to analyze, as it prevents a further time-scale separation between the primal and the dual dynamics (see, e.g., [2]), there are cases where different stepsize values are preferable (for instance, if one wants to embed the multiplication by n of $\nabla \ell_i(x_i^t, \lambda_i^t)$ in (2) within the stepsize bound). The analysis carried out here can be easily extended to cover such a case.

1.4 Notation

If $\sim$ is a binary relation on a set S and $z \in S$, we let $S_{\sim z} := \{s \in S : s \sim z\}$. If not otherwise specified, binary relations are applied component-wise to vectors. We denote by ∇f the gradient of a differentiable function f ; $\sigma(A)$ denotes the spectrum of a matrix $A \in \mathbb{R}^{n \times n}$; A is Schur if $\sigma(A)$ is contained in the open unit disk in the complex plane; $A > 0$ means that A is positive definite. If $\mathcal{J}$ is an ordered set of finite cardinality n , $(x_i)_{i \in \mathcal{J}}$ denotes the n -tuple of $\mathbb{R}^n$ ordered by $\mathcal{J}$ in the obvious way. Set inclusion (either strict or not) is denoted by $\subset$. With $n \in \mathbb{N}$, we let $\mathbf{1} := (1, \dots, 1) \in \mathbb{R}^n$, $I \in \mathbb{R}^{n \times n}$ be the identity matrix, and $\langle x, y \rangle := \sum_{h=1}^n x_h y_h$ denote the standard inner product of two vectors $x, y \in \mathbb{R}^n$. Moreover, $|\cdot|$ denotes the Euclidean norm of a vector or the matrix-induced 2-norm, and the closed ball of radius r around point $\bar{x} \in \mathbb{R}^n$ is denoted by $\overline{\mathbb{B}}_r(\bar{x}) := \{x \in \mathbb{R}^n : |x - \bar{x}| \leq r\}$.

We let $S \in \mathbb{R}^{n \times (n-1)}$ be a matrix satisfying

$$S^\top \mathbf{1} = 0, \quad S^\top S = I_{n-1}. \quad (3)$$

Moreover, we define the matrix $T \in \mathbb{R}^{n \times n}$ and its inverse as

$$T := \begin{bmatrix} \frac{\mathbf{1}^\top}{n} \\ S^\top \end{bmatrix}, \quad T^{-1} := [\mathbf{1} \ S]. \quad (4)$$

From (3)-(4), we deduce that

$$I_n = \frac{1}{n} \mathbf{1} \mathbf{1}^\top + S S^\top, \quad |S| = 1. \quad (5)$$